



General Description

With over 15 years in the European and US markets, Greenlight Pro (4mm) is a trusted solution for integrating photovoltaic (PV) systems with green roof assemblies.

Ballasted, Penetration-Free Design

Greenlight Pro is secured in place by roofing overburden—such as green roof media or stone aggregate—eliminating the need for welded or bolted connections to the roof structure. This design greatly reduces waterproofing penetrations and flashing requirements compared to traditional systems.

Flexible Installation

The system's HDPE base plate can bend and conform to roof slopes of up to 2% (and up to 5° overall pitch tolerance), making it ideal for new or existing roofs with slight variations or undulations.

Integrated Conduit Management

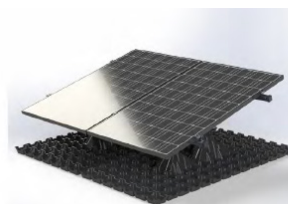
All electrical conduits and wiring can be run underneath the system within the green roof profile, simplifying maintenance and improving aesthetics.

Material Properties

Base Plate:	High-Density Polyethylene (HDPE)
Dimensions:	78" × 39" × 2.5"
Hardware:	Magnelis®, Aluminum, and Stainless Steel

Technical Data

- Weight: 26.5 lbs. per unit (not incl. PV panels and ballast)
- Maximum Roof Slope: 5°
- Orientation & Tilt Options: Portrait or Landscape orientations at standard inclinations of 10°, 15°, or 20°.
- Water Storage Capacity: 10.5 gal. / 1.4 cubic feet per panel.
- Ballast Requirements: Determined on a per-project basis, guided by wind uplift and layout studies.
- Roof Compatibility: Suitable for new or existing roofs, with multiple layout configurations.
- Minimum Soil Clearance: Most regions require a 12" clearance between the top of soil and the lowest point of the PV panel.
- Complete unit includes: one (1) base plate, one (1) angled bracket, two (2) profile rails (length vary) and hardware.
- PV panels, electrical components and wiring not included.

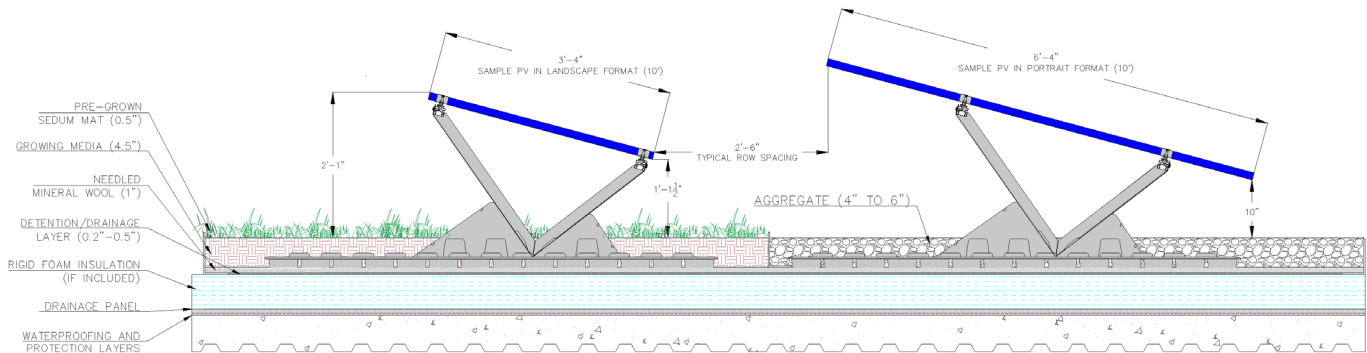


shown with pv panel in portrait mode



shown with pv panel in landscape mode

This data is provided for informational purposes only. This document is not a warranty as to the suitability or the fitness for a specific use or merchantability of products referred to, no guarantee of satisfactory results from reliance upon contained information or recommendation and disclaims all liability from resulting loss or damage. This information is subject to change without notice.



Delivery & Installation

Comprehensive Roof Plan

Installers receive a detailed layout indicating the exact position and spacing of each base plate.

Complete Material Shipment

All necessary components—including base plates, angled brackets, profile rails, clips, and endcaps—are delivered to the jobsite, with extra parts to cover typical waste factors.

Convenient Scope of Work

Roofing contractors or green roof landscapers can easily include the installation of this system in their overall project scope, streamlining coordination and scheduling.

Additional Advantages

Reduced Structural Impact

By eliminating penetrations and flashing details, Greenlight Pro by Contec reduces long-term maintenance and potential points of water ingress.

Design Flexibility

The V-shaped brackets are custom-cut to match the desired PV panel angles, roof slope, and soil depth. Knowing the exact PV panel item number prior to design ensures the most accurate fit.

Improved Aesthetics & Function

Combining green roof and solar technologies enhances rooftop aesthetics, maximizes available space, and can improve cooling efficiency by reducing ambient roof temperature.

Application

PV Panel Angle

A 10° angle is typically preferred, as it requires standard ballast levels that most green roofs can accommodate. Steeper angles (15°, 20°) are possible but may require additional ballast.

Ballast & Panel Orientation

Landscape Orientation: Generally requires 17–24 lbs/ft² of ballast. The center of the array needs less ballast, while perimeter edges and corners need more.

Portrait Orientation: Typically requires slightly higher ballast loads overall.

Wind Uplift Analysis

A detailed wind uplift study is provided by Sempergreen to recommend ballast requirements (lbs./sf) per roof area. The U.S. market requires a minimum 4 mm thick base plate.

Soil-Based Securement

The system relies on the weight of the green roof soil or other overburden (e.g., stone aggregate) to remain firmly in place.